

Economic Growth and CO₂ Emissions in Asia:

A Dynamic Panel Analysis of the Environmental Kuznets Curve

Abstract

This study examines the relationship between economic development and carbon dioxide (CO₂) emissions per capita for a balanced panel of 16 Asian countries over 2000–2023. Data were collected and cleaned in Python using World Bank and Our World in Data sources and analyzed in Stata. The empirical strategy combines descriptive analysis, panel unit-root tests, panel cointegration investigation, dynamic Panel ARDL estimation using the Pooled Mean Group (PMG) estimator, static FE/RE benchmarks, and dependence-robust inference. The core quadratic PMG model produces a positive coefficient on log GDP per capita and a negative coefficient on its squared term, but its implied turning point lies far outside the observed income range. The full PMG specification preserves the positive/negative quadratic pattern and implies a turning point of approximately \$15,406 in constant 2015 US dollars, within the sample income range. Diagnostics reveal both serial correlation and cross-sectional dependence. A Driscoll–Kraay fixed-effects specification is therefore used as a robustness check. The results are interpreted as evidence of a downward-opening quadratic relationship in the fitted specifications, rather than proof that every country follows an identical EKC transition.

1 Introduction

The Environmental Kuznets Curve (EKC) hypothesis proposes that the relationship between economic development and environmental degradation may be nonlinear. At relatively low income levels, economic expansion may be associated with increasing environmental pressure, while at sufficiently high income levels the relationship may weaken or reverse. A quadratic EKC specification represents this possibility through a positive coefficient on income and a negative coefficient on income squared.

This study examines the income–CO₂ relationship for a diverse group of Asian economies over 2000–2023. The analysis goes beyond a simple static regression by examining the time-series properties of the panel and using a dynamic Panel ARDL framework. The objectives are to construct and validate a balanced panel dataset, examine integration and long-run relationships, estimate short-run and long-run income–emissions effects, and assess whether the central quadratic relationship survives alternative static and dependence-robust specifications.

2 Data Collection and Preparation

2.1 Sample

The final sample contains 16 Asian countries observed annually from 2000 to 2023, giving

$$16 \times 24 = 384$$

country-year observations. The countries are:

Table 1: Final country sample	
Region	Countries
South Asia	India, Pakistan, Bangladesh, Nepal
Southeast Asia	Indonesia, Malaysia, Philippines, Thailand, Vietnam
East Asia	China, Japan, South Korea, Mongolia
West/Central Asia	Türkiye, Kazakhstan, Saudi Arabia

Sri Lanka was excluded because several trade-related indicators contained a five-year gap during 2010–2014. The exclusion preserved a strongly balanced panel.

2.2 Data Sources and Variables

The dataset was collected and prepared in Python using Google Colab. The main sources were the World Bank’s World Development Indicators (WDI) and Our World in Data (OWID).

Table 2: Variables and sources

Variable	Source	Definition
GDP per capita	World Bank	Constant 2015 US dollars
CO ₂ emissions per capita	OWID	CO ₂ tonnes per person
Trade	World Bank	Trade, percent of GDP
Population growth	World Bank	Annual percent growth
Urban population	World Bank	Percent of total population
Industry	World Bank	Industry incl. construction, percent of GDP
Renewable energy	OWID	Percent of final energy consumption
Fossil fuels	OWID	Fossil-fuel consumption per capita

World Bank indicators were GDP per capita (`NY.GDP.PCAP.KD`), trade (`NE.TRD.GNFS.ZS`), population growth (`SP.POP.GROW`), urban population (`SP.URB.TOTL.IN.ZS`), and industry including construction (`NV.IND.TOTL.ZS`). OWID supplied CO₂ per capita, renewable energy consumption, and fossil-fuel consumption.

During cleaning, alternative sources were compared. The World Bank and OWID renewable-energy series differed relatively strongly for Indonesia, so OWID was used consistently. The World Bank fossil-fuel series also contained missing or unusual observations, so the OWID series was used in the final dataset. Data were reshaped, filtered, merged using country code and year, and checked for missing values, duplicate country-year observations, coverage, and balance. The final dataset contains 384 observations with no missing values or duplicate country-year combinations.

3 Descriptive Analysis

3.1 Descriptive Statistics

Table 3: Descriptive statistics

Variable	Mean	Std. Dev.	Min.	Max.
GDP per capita	8702.51	10214.39	547.22	37902.55
CO ₂ per capita	5.76	5.78	0.09	23.24
Trade	73.27	42.51	19.28	220.41
Population growth	1.24	1.02	-2.55	5.41
Urban population	56.09	19.92	23.31	92.08
Industry	33.31	9.79	11.79	66.43
Renewable energy	22.72	22.24	0.01	91.31
Fossil fuels	21404.79	24152.81	339.54	103513.90

3.2 Correlation Analysis

Log GDP per capita and log CO₂ emissions per capita have a strong positive correlation, $r = 0.888$ with $p < 0.001$. GDP is also strongly correlated with urban population ($r = 0.837$), renewable energy consumption ($r = -0.814$), and fossil-fuel consumption ($r = 0.809$). CO₂ emissions correlate positively with fossil fuels ($r = 0.775$), urban population ($r = 0.702$), and industry ($r = 0.618$), and negatively with renewable energy ($r = -0.916$).

The correlation matrix is descriptive and does not establish causality or an EKC. A preliminary scatterplot showed a strong positive association between log GDP and log CO₂, with some nonlinearity, but the plot alone does not establish an inverted-U relationship.

4 Variable Transformation and Panel Setup

The dependent variable is

$$\ln CO2_{it} = \ln(\text{CO}_2 \text{ emissions per capita}_{it}).$$

Economic development is represented by

$$\ln GDP_{it} = \ln(\text{GDP per capita}_{it}),$$

and the quadratic term is

$$(\ln GDP_{it})^2.$$

The main Stata variables are `ln_co2`, `ln_gdp`, and `ln_gdp2`. Controls are retained in levels because population growth can be negative and percentage/share variables are naturally interpretable in levels.

The panel was declared using `xtset country_id year`. The resulting panel is strongly balanced with 16 countries and 24 annual observations per country.

5 Panel Unit-Root Tests

Fisher-ADF and Im-Pesaran-Shin (IPS) tests were used. The Fisher unit-root null is that all panels contain unit roots, while the alternative permits at least one panel to be stationary. The results classify the variables as I(0) or I(1), with no evidence of I(2) behavior.

Table 4: Integration classification

Variable	Level	First difference
$\ln CO_2$	I(1)	Stationary
$\ln GDP$	I(1)	Stationary
Trade	I(0)	Already stationary
Population growth	I(1)	Stationary
Urban population	I(0)	Already stationary
Industry	I(1)	Stationary
Renewable energy	I(1)	Stationary
Fossil fuels	I(1)	Stationary

The mixture of I(0) and I(1) variables, with no I(2) variables, is compatible with a Panel ARDL/PMG framework.

6 Panel Cointegration Investigation

The Hadri-Kurozumi-Rao (H-K-R) panel cointegration test was used. Its null hypothesis is panel cointegration:

$$H_0 : \text{all units are cointegrated,}$$

against

$$H_1 : \text{at least one unit is not cointegrated.}$$

For the core linear relationship between $\log \text{CO}_2$ and $\log \text{GDP}$, the bias-corrected DOLS statistic was

$$\tilde{S}_K = -0.1705, \quad p = 0.5677,$$

so the cointegration null was not rejected. A quadratic core specification also failed to reject the cointegration null under the H-K-R procedure. However, the energy-extended specification rejected the cointegration null ($p = 0.0001$), as did the full I(1)-control specification ($p < 0.001$).

These results indicate that evidence for panel-wide cointegration is sensitive to the variable set and test specification. They do not mean that the variables have no economic relationship. The PMG results are therefore interpreted as dynamic estimates under the specified Panel ARDL framework, rather than as proof of a universal cointegrating vector.

7 Dynamic Panel ARDL and PMG

7.1 General Specification

The quadratic long-run relationship is represented by

$$\ln \text{CO2}_{it} = \alpha_i + \beta_1 \ln \text{GDP}_{it} + \beta_2 (\ln \text{GDP}_{it})^2 + \boldsymbol{\gamma}' \mathbf{Z}_{it} + \varepsilon_{it},$$

where $\mathbf{Z}_{it}$ contains the I(1) long-run controls.

PMG imposes common long-run coefficients across countries while allowing short-run coefficients and adjustment speeds to differ across countries. The error-correction coefficient measures adjustment toward the estimated long-run relationship.

7.2 Core Quadratic PMG

The core long-run specification is

$$\ln \text{CO2}_{it} = \alpha_i + \beta_1 \ln \text{GDP}_{it} + \beta_2 (\ln \text{GDP}_{it})^2 + \varepsilon_{it}.$$

The estimated coefficients are

$$\hat{\beta}_1 = 1.582959, \quad p < 0.001,$$

$$\hat{\beta}_2 = -0.0598304, \quad p = 0.002.$$

The error-correction coefficient is

$$\hat{\phi} = -0.2902, \quad p < 0.001.$$

Short-run changes in log GDP and squared log GDP are statistically insignificant ($p = 0.961$ and $p = 0.963$).

The MG alternative gives $\hat{\beta}_{1,MG} = 35.9919$ ($p = 0.070$) and $\hat{\beta}_{2,MG} = -1.7931$ ($p = 0.062$). The MG–PMG Hausman test is

$$\chi^2(2) = 3.29, \quad p = 0.1930.$$

The null of no systematic difference is not rejected, providing methodological support for the common long-run restriction in the core specification.

7.3 Core Turning Point

For

$$y = \beta_1 x + \beta_2 x^2,$$

the turning point is

$$x^* = -\frac{\beta_1}{2\beta_2}.$$

Using the core PMG coefficients,

$$\ln GDP^* = -\frac{1.582959}{2(-0.0598304)} = 13.2287,$$

and

$$GDP^* = e^{13.2287} \approx \$556,108.$$

The maximum GDP per capita in the sample is approximately \$37,903. Therefore the core turning point lies far outside the observed income range. The core coefficients have the downward-opening quadratic sign pattern, but the model does not show an observed inverted-U transition within the sample.

8 Energy-Extended PMG

The energy model is

$$\ln CO2_{it} = \beta_1 \ln GDP_{it} + \beta_2 (\ln GDP_{it})^2 + \beta_3 Renewable_{it} + \beta_4 Fossil_{it} + \varepsilon_{it}.$$

The PMG estimates are

$$\begin{aligned} \hat{\beta}_1 &= 5.276918, & p &< 0.001, \\ \hat{\beta}_2 &= -0.2744128, & p &< 0.001, \\ \hat{\beta}_3 &= 0.0002593, & p &= 0.942, \end{aligned}$$

$$\widehat{\beta}_4 = 0.0000251, \quad p < 0.001.$$

The error-correction coefficient is -0.2998673 ($p < 0.001$). In the short run, the change in fossil-fuel consumption is positive and significant (0.0000969 , $p = 0.005$), while short-run changes in GDP, squared GDP, and renewable energy are insignificant.

The MG–PMG Hausman test gives

$$\chi^2(3) = 0.77, \quad p = 0.8562,$$

so the null of no systematic difference is not rejected.

The energy turning point is

$$\ln GDP^* = -\frac{5.276918}{2(-0.2744128)} = 9.6149,$$

with

$$GDP^* = e^{9.6149} \approx \$14,987.$$

This lies within the observed income range. It is the implied peak of the fitted quadratic function and is not, by itself, proof that every country follows an EKC transition.

9 Full PMG Specification

9.1 Specification

The full long-run variables are log GDP, squared log GDP, population growth, industry, renewable energy, and fossil fuels. Trade and urban population are $I(0)$ and enter as short-run controls.

The long-run equation is

$$\begin{aligned} \ln CO2_{it} = & \beta_1 \ln GDP_{it} + \beta_2 (\ln GDP_{it})^2 \\ & + \beta_3 PopulationGrowth_{it} + \beta_4 Industry_{it} \\ & + \beta_5 Renewable_{it} + \beta_6 Fossil_{it} + \varepsilon_{it}. \end{aligned}$$

The full PMG converged successfully after 44 iterations. The long-run estimates are:

Table 5: Full PMG long-run estimates

Variable	Coefficient	<i>p</i> -value
$\ln GDP$	1.8015	< 0.001
$(\ln GDP)^2$	-0.0934	< 0.001
Population growth	-0.0713	< 0.001
Industry	0.0080	< 0.001
Renewable energy	-0.0116	< 0.001
Fossil fuels	0.0000272	< 0.001

The error-correction coefficient is

$$\hat{\phi} = -0.3786, \quad p < 0.001,$$

indicating statistically significant adjustment toward the estimated long-run relationship. Approximately 37.9% of the previous period's estimated disequilibrium is corrected per period.

The short-run change in fossil-fuel consumption is positive and significant:

$$\widehat{\Delta Fossil} = 0.0001044, \quad p = 0.010.$$

The short-run coefficients on GDP, squared GDP, population growth, industry, renewable energy, trade, and urban population are not statistically significant at the 5% level.

9.2 Full Turning Point

The full PMG turning point is

$$\ln GDP^* = -\frac{1.80154}{2(-0.0934165)} = 9.642515,$$

and therefore

$$GDP^* = e^{9.642515} = \$15,406.04.$$

The observed GDP-per-capita maximum is approximately \$37,903, so the implied turning point lies within the observed income range. It represents the peak of the fitted common long-run quadratic relationship, not proof of an identical country-level EKC trajectory.

9.3 Full MG Limitation

A full MG model was attempted using the same long-run variables, but it did not converge under the available `xtpmg` implementation. The `difficult` option was also attempted

without resolving the problem. Therefore a valid full-model MG–PMG Hausman test cannot be reported. The specification was not simplified merely to obtain convergence. The full PMG was retained based on successful estimation, theoretical interest in a common long-run relationship, and supporting evidence from the energy-model Hausman test.

10 Static FE/RE Benchmark

The static benchmark is

$$\ln CO2_{it} = \alpha_i + \beta_1 \ln GDP_{it} + \beta_2 (\ln GDP_{it})^2 + \boldsymbol{\gamma}' \mathbf{X}_{it} + \varepsilon_{it}.$$

The FE estimates are

$$\begin{aligned} \hat{\beta}_{GDP} &= 4.4474, & p < 0.001, \\ \hat{\beta}_{GDP^2} &= -0.2418, & p < 0.001, \end{aligned}$$

with within $R^2 = 0.8333$. The RE estimates are similar in direction and magnitude. The FE–RE Hausman test gives

$$\chi^2(7) = 7.03, \quad p = 0.4254.$$

The null of no systematic difference is not rejected. FE and RE are treated as benchmarks, while PMG remains the main dynamic specification.

11 Multicollinearity Diagnostic

VIFs were calculated using a static regression containing the explanatory variables from the full specification. This was a design-matrix diagnostic, not the main model.

Table 6: Variance Inflation Factors

Variable	VIF
$\ln GDP$	646.75
$(\ln GDP)^2$	632.56
Fossil fuels	6.71
Renewable energy	5.99
Urban population	4.20
Industry	3.18
Population growth	1.56
Trade	1.42
Mean VIF	162.80

The very high VIFs for GDP and GDP squared are expected because the quadratic term is mechanically constructed from GDP. Both terms are required for the quadratic specification and were not removed solely because of VIF. Among controls, fossil fuels and renewable energy show moderate multicollinearity. No control was removed solely on the basis of VIF.

12 Cross-Sectional Dependence

Pesaran’s CD test was applied to the residuals of the static FE benchmark. The result was

$$CD = 2.119, \quad p = 0.0341.$$

The null of cross-sectional independence is rejected at 5%. The average absolute off-diagonal residual correlation is 0.436, indicating non-negligible cross-country dependence. This suggests that common shocks or correlated regional/global factors may affect countries simultaneously.

13 Serial Correlation

The Wooldridge test for autocorrelation in panel data tests

$$H_0 : \text{no first-order autocorrelation}$$

against

$$H_1 : \text{first-order autocorrelation exists.}$$

The result is

$$F(1, 15) = 27.543, \quad p = 0.0001.$$

The null is rejected, indicating strong evidence of first-order serial correlation in the panel errors. In intuitive terms, unexplained movements in CO₂ within a country are correlated over time.

14 Clustered FE and Driscoll–Kraay Robustness

A country-clustered FE model allows errors within a country to be correlated over time. The point estimates remain unchanged because clustering changes the variance-covariance matrix, not the coefficients. The central estimates remain significant:

$$\hat{\beta}_{GDP} = 4.4474, \quad p = 0.021,$$

$$\hat{\beta}_{GDP^2} = -0.2418, \quad p = 0.028.$$

Because both serial correlation and cross-sectional dependence were detected, a fixed-effects model with Driscoll–Kraay standard errors was then estimated. Driscoll–Kraay changes the inference while leaving the FE coefficient estimates unchanged. The results are

$$\hat{\beta}_{GDP} = 4.4474, \quad p < 0.001,$$

$$\hat{\beta}_{GDP^2} = -0.2418, \quad p < 0.001.$$

The central quadratic relationship therefore remains statistically significant under dependence-robust inference. The `xtscc` output reported a maximum lag of 2.

15 Integrated Interpretation

The empirical strategy should be read as a sequence. The unit-root tests establish that the variables are I(0) or I(1), with no I(2) evidence. The core H-K-R results are consistent with a long-run relationship, while expanded cointegration tests are more sensitive to the added variables. The PMG framework then separates short-run adjustment from the estimated common long-run relationship.

The core PMG model has the expected positive/negative quadratic signs, but its turning point of approximately \$556,108 is far beyond the observed income range. The energy model produces an implied turning point of approximately \$14,987, while the full PMG model produces approximately \$15,406. The full model preserves the positive GDP and negative GDP-squared coefficients after adding demographic, structural, and energy controls.

The static FE benchmark and the Driscoll–Kraay specification also preserve the positive GDP and negative GDP-squared pattern. The panel nevertheless exhibits serial correlation and cross-sectional dependence, which is why the dependence-robust specification is useful as a benchmark robustness check.

The overall evidence is therefore consistent with a downward-opening quadratic relationship in the fitted models. However, this should not be presented as proof that every country experienced an EKC transition. PMG imposes common long-run coefficients, and the turning point represents the peak of the fitted common relationship.

16 Main Findings

1. The final dataset contains 16 Asian countries and 384 balanced observations over 2000–2023.
2. The main variables are $I(0)$ or $I(1)$, with no evidence of $I(2)$.
3. The core PMG model has positive and significant GDP and negative and significant GDP-squared coefficients.
4. The core PMG turning point is approximately \$556,108, outside the observed income range.
5. The energy-extended PMG turning point is approximately \$14,987 and lies within the observed range.
6. The full PMG turning point is approximately \$15,406 and lies within the observed range.
7. The full PMG error-correction coefficient is approximately -0.379 , indicating significant adjustment toward the estimated long-run relationship.
8. The full MG model did not converge under the available `xtpmg` implementation, so a full MG–PMG Hausman test is unavailable.
9. Static FE and Driscoll–Kraay FE preserve the central positive/negative quadratic coefficient pattern.
10. The panel exhibits both serial correlation and cross-sectional dependence.

17 Limitations

The panel has 16 countries and 24 annual observations per country, so the time dimension remains relatively short for some panel time-series procedures. The H-K-R cointegration

evidence is sensitive to the variable set. The PMG estimator imposes common long-run coefficients, and the full MG model could not be estimated successfully. Finally, CO₂ emissions per capita are only one measure of environmental pressure, and other environmental indicators could yield different relationships.

18 Analyses Not Retained

18.1 Cubic PMG

A cubic PMG specification was considered but was not retained because the final project focuses on the quadratic EKC and a cubic PMG would require an additional full dynamic estimation and diagnostic workflow. A cubic specification is therefore not part of the final main analysis.

18.2 Exports and Imports

Exports and imports were initially collected as candidate controls but were not retained in the final master dataset. The broader trade measure was retained instead.

18.3 Sri Lanka

Sri Lanka was excluded because of the five-year gap in relevant trade-related indicators during 2010–2014. This preserved the strongly balanced panel.

19 Conclusion

This study constructs a balanced panel of 16 Asian economies over 2000–2023 and examines the relationship between economic development and CO₂ emissions using dynamic and static panel methods. The PMG estimates consistently produce a positive coefficient on log GDP and a negative coefficient on squared log GDP. The core model’s turning point is far outside the sample, whereas the energy-extended and full models imply turning points within the observed income range.

The full PMG model also indicates a negative long-run coefficient on renewable energy and a positive coefficient on fossil-fuel consumption, alongside significant error correction. Static FE and Driscoll–Kraay FE estimates preserve the central quadratic coefficient pattern. At the same time, serial correlation and cross-sectional dependence are present in the panel and are explicitly considered in the robustness analysis.

The findings are best described as evidence of a downward-opening quadratic relationship in the fitted specifications, with the full model implying an income turning point around \$15,406. The turning point should be understood as a feature of the estimated

common long-run function rather than direct evidence that every country in the sample has followed the same EKC trajectory.

20 Exploratory Visualization and Country-Level Patterns

To complement the formal panel-data analysis, exploratory visualizations were constructed in Tableau using the cleaned country-year dataset. The visualizations provide an intuitive representation of the relationship between economic development and carbon dioxide emissions before accounting for the additional controls and dynamic structure incorporated in the econometric models.

The main scatter plot presents GDP per capita against CO₂ emissions per capita for the 16 countries over the 2000–2023 period. A corresponding logarithmic visualization was also constructed using the natural logarithms of GDP per capita and CO₂ emissions per capita. Each observation represents a country-year pair, resulting in 384 observations. Country identifiers and year information were retained to allow the evolution of individual countries to be examined.

The visualizations are intended primarily as descriptive and exploratory tools. In particular, the raw scatter plots do not control for population growth, industrial structure, renewable energy consumption, fossil-fuel consumption, trade, or urbanization. They also do not incorporate the country-specific short-run dynamics and common long-run restrictions imposed by the PMG estimator. Consequently, visual patterns in the Tableau figures should not be interpreted as direct estimates of the EKC relationship.

20.1 Country-Specific Visual Patterns

Country-specific polynomial trend lines were examined as an additional exploratory exercise. This exercise revealed substantial heterogeneity in the shapes of the raw GDP–CO₂ relationship across countries. In particular, the fitted polynomial curves displayed a U-shaped pattern for Indonesia, Mongolia, Nepal, Pakistan, and the Philippines. By contrast, India and Vietnam displayed predominantly upward-sloping patterns in the corresponding visualization.

These differences are important because they demonstrate that the country-level raw data do not necessarily exhibit a common inverted-U pattern. Some countries appear to move through portions of the income–emissions relationship that are consistent with rising emissions over the observed income range, while others display a fitted curvature in the opposite direction. The existence of such heterogeneous visual patterns is not inconsistent with the PMG results.

The PMG estimator imposes homogeneity on the long-run coefficients while allowing short-run coefficients and adjustment dynamics to differ across countries. The Tableau plots, in contrast, show the unadjusted country-year observations. They therefore capture a different aspect of the data. A country's observed trajectory can be affected by changes in energy composition, industrial structure, population growth, trade, urbanization, technological change, and other factors that are controlled for in the econometric specification. In addition, the observed trajectory may cover only part of the underlying long-run relationship.

Thus, a U-shaped country-specific polynomial trend in the descriptive visualization does not imply that the country has a statistically significant U-shaped structural relationship. Similarly, a visually upward-sloping trajectory does not by itself establish that the country does not exhibit an EKC after controlling for the other variables in the model.

20.2 Why the Visual Patterns Differ from the PMG Specification

An important distinction must therefore be made between the descriptive country-level trajectories and the estimated panel relationship. The quadratic PMG specification is based on the long-run equation

$$\ln(CO_{2,it}) = \alpha_i + \beta_1 \ln(GDP_{it}) + \beta_2 [\ln(GDP_{it})]^2 + \boldsymbol{\gamma}' \mathbf{X}_{it} + u_{it}, \quad (1)$$

where $\mathbf{X}_{it}$ contains the relevant control variables. The PMG framework further permits the short-run coefficients and the speed of adjustment to differ across countries.

The Tableau visualization does not estimate this conditional relationship. Instead, it plots the observed values of GDP per capita and CO₂ emissions per capita. Consequently, the visual shape for a particular country can differ from the common long-run quadratic relationship estimated by the PMG model.

This distinction is particularly relevant for countries such as Indonesia, Mongolia, Nepal, Pakistan, and the Philippines, for which the exploratory polynomial visualization produces a U-shaped fitted curve. Such a pattern should be interpreted as evidence of heterogeneity in the raw observed relationship rather than as evidence against the PMG specification. The visualization and the econometric model answer different questions: the former describes the observed data, whereas the latter estimates a conditional long-run relationship after accounting for the variables included in the model and the dynamic structure of the panel.

20.3 Role of Additional Variables in Visualization

The final econometric specification incorporates several additional variables, including population growth, industrial structure, renewable energy consumption, fossil-fuel consumption, trade, and urbanization. These variables cannot all be represented simultaneously in a conventional two-dimensional GDP–CO₂ scatter plot without substantially changing the nature of the visualization.

Accordingly, the Tableau analysis focuses on the principal variables of interest, GDP per capita and CO₂ emissions per capita, while the additional variables are incorporated formally through the econometric models. Separate time-series and country-comparison visualizations are used to provide additional descriptive information on the evolution of economic activity and emissions across the sample.

This separation between visualization and econometric estimation is deliberate. The Tableau dashboard provides an accessible overview of the data and highlights cross-country heterogeneity, whereas the Stata analysis provides the formal statistical framework for evaluating the long-run relationship.

20.4 Interpretation of Polynomial Trend Lines

The country-specific polynomial trend lines should be treated as exploratory rather than inferential. A polynomial curve can summarize the shape of a cloud of observations, but its curvature does not establish statistical significance, causality, or the presence of an EKC turning point. In particular, the apparent U-shaped patterns observed for several countries may arise because the available observations cover only a particular range of the income distribution or because other determinants of emissions vary simultaneously with income.

For this reason, the polynomial trend lines are not used to replace the formal PMG estimates. Instead, they are presented as a complementary visual diagnostic that illustrates the heterogeneity of the raw data and motivates the use of a panel framework rather than relying exclusively on a simple visual interpretation of the GDP–CO₂ relationship.

References

- Akpan, U. F. and Abang, D. E. (2015). “Environmental quality and economic growth: A panel analysis of the ‘U’ in Kuznets.” *Journal of Economic Research*, 20, 317–339.
- World Bank. *World Development Indicators*. Indicators used include GDP per capita, trade, population growth, urban population, and industry value added.

- Our World in Data. Datasets on CO₂ emissions per capita, renewable energy consumption, and fossil-fuel consumption per capita.

A Core Stata Commands

A.1 Core PMG

```
xtpmg d.ln_co2 d.ln_gdp d.ln_gdp2, ///
      lr(1.ln_co2 ln_gdp ln_gdp2) ///
      pmg ec(ect_quad)
```

A.2 Energy PMG

```
xtpmg d.ln_co2 d.ln_gdp d.ln_gdp2 ///
      d.renewable_energy_consumption d.fossil_fuels, ///
      lr(1.ln_co2 ln_gdp ln_gdp2 ///
      renewable_energy_consumption fossil_fuels) ///
      pmg ec(ect_energy)
```

A.3 Full PMG

```
xtpmg d.ln_co2 d.ln_gdp d.ln_gdp2 ///
      d.population_growth d.industry ///
      d.renewable_energy_consumption d.fossil_fuels ///
      d.trade d.urban_population, ///
      lr(1.ln_co2 ln_gdp ln_gdp2 ///
      population_growth industry ///
      renewable_energy_consumption fossil_fuels) ///
      pmg ec(ect_full)
```

A.4 Static FE and Robustness

```
xtreg ln_co2 ln_gdp ln_gdp2 ///
      population_growth industry ///
      renewable_energy_consumption fossil_fuels ///
      trade urban_population, fe
```

```
xtreg ln_co2 ln_gdp ln_gdp2 ///
      population_growth industry ///
      renewable_energy_consumption fossil_fuels ///
```

```

trade urban_population, fe vce(cluster country_id)

xtcsd, pesaran abs

xtserial ln_co2 ln_gdp ln_gdp2 ///
population_growth industry ///
renewable_energy_consumption fossil_fuels ///
trade urban_population

xtscc ln_co2 ln_gdp ln_gdp2 ///
population_growth industry ///
renewable_energy_consumption fossil_fuels ///
trade urban_population, fe

```